Assignment 2: Firewalls

Deadline: Friday 2/10 17:00

This assignment can be done by at most two persons. Email your solution to Elmira.

Name1: _____

Name2: _____

Firewalls

This section contains questions related to security issues of firewalls; how this technology works and the different types of firewalls.

1. What are the two characteristics of static packet filter firewalls?

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2. How do stateful firewalls work for TCP?

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3. Can stateful firewalls maintain state information for connectionless protocols like UDP and ICMP ?

   …

4. How does a NAT firewall work?

   …

5. Explain application firewall operation.

   …

5. If you will proxy four applications, how many proxy programs will you need?

   …

7. Should the last rule of a screening firewall be Deny All or Permit All? Explain.

   …

8. You have a rule in your ACL to block a particular type of traffic. However, when you do an audit, you find that the firewall is not blocking this traffic. What is the problem likely to be?

…

9. Assume that LiU has decided to have the following security policy:

- to allow only incoming (inbound) connections to port 25 of LiU mail server (130.236.8.134)

- to allow all outbound connections, that is, to permit all connections initiated by LiU internal hosts.

The LiU technician Pier implements the following ACL:

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allow tcp * : * → 130.236.8.134 : 25

drop tcp * : * → 130.236.8.134 : *

allow tcp 130.236.136.96 : * → * : *

allow tcp 130.236.136.97 : * → * : *

. . . // plus one such a rule for every internal IP address

drop * * : * → * : *

Does this rule set enforce the LiU security policy? Explain.

This one is a tricky question, be careful.

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